Phone Hseng

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Education

B.S. in Computer Science Aug 2026 – May 2028
Virginia Tech *Blacksburg, VA*
Relevant Coursework: Software Design and Data Structures, Intro to Software Design, Intro to Problem Solving in CS, Intro to Computer Organization I, Foundations of Engineering Practice, Calculus I, Calculus II

Projects

Clear Home | *Python, Flask, Gemma 4, Google Cloud Run, Google Maps* clearhome.live | github.com/phonehseng/clearhome
- **Won 1st place overall and 1st place for Google Cloud usage** at GDG Build with AI × Hacktoberfest 2026 with a team of four, building an AI housing assistant that finds verified affordable homes, shelters, and rent help
- Served Gemma 4 from our own Cloud Run L4 GPU with a Gemini API fallback, streaming answers grounded only in a hand-verified list of 400+ records so the assistant cannot invent listings
- Added map search and measured walking/transit times with Google Maps APIs, plus a daily job that finds new listings, checks their sources, and redeploys

Letters | *Rust, PDFium, Tauri, Svelte, TypeScript* github.com/phonehseng/Letters
- Building a cross-platform desktop PDF editor and office suite for Windows and macOS, structured as 20+ Rust crates (rendering, text editing, conversion, forms, OCR, search) behind a Tauri and Svelte interface
- Wrote an exact PDF-to-Word converter that keeps every line in place to within 1/100 of a point on 12 of 13 test documents, and in-place editing of a PDF's own text in its embedded fonts; pages render as prioritized tiles so large files stay responsive

Self-Play Reinforcement Learning Lab | *JavaScript, Web Workers, Python* Private repository
- Implemented a neural network and backpropagation from scratch (no ML library) that learns chess purely from self-play with TD(λ), training in a Web Worker with configurable layers, learning rate, and exploration
- Probed the trained network by measuring evaluation drops when pieces are removed; learned piece values converge toward the classical 1/3/3/5/9 without being given them

DisplayFlip | *C++, Win32 API* github.com/phonehseng/DisplayFlip
- Built a native Windows display-switching utility (~770 KB executable, ~2 MB RAM) adding ~20 ms over the OS mode switch; identifies monitors by EDID and restores window layouts and audio routing per display profile

Atlas | *Python, Go, JavaScript, Docker* github.com/phonehseng/atlas
- Built a self-hosted media platform and guided installer that deploys and connects about ten services (Jellyfin, Sonarr, Radarr, VPN, Tailscale) natively on Windows or as Docker containers on Linux, with health checks that diagnose and repair failures

Experience

Service Technician Apr 2026 – Aug 2026
Micro Center *Fairfax, VA*
- Diagnosed and repaired customer desktop PCs, laptops, and Apple devices, performing board-level diagnostics, hardware/software testing, performance and stability testing, and GPU/driver troubleshooting
- Resolved 12+ repair tickets per day end-to-end, from intake and diagnosis through repair, testing, and customer handoff
- Earned vendor-authorized repair certifications from Apple, HP, Dell, Lenovo (NGSP/DOA), and ASUS

Customer Service Representative Aug 2025 – Apr 2026
Micro Center *Fairfax, VA*
- Processed 20+ transactions per hour (chain average: 16.5) while maintaining a 91% customer capture rate across PC components, prebuilt systems, and peripherals
- Ranked in the top 100 warranty sellers chainwide, attaching service plans on 13% of transactions vs. a 6.8% chain average and generating $5.03 in plan revenue per customer (chain average: $2.70)
- Received the store's GEAR Award (Employee of the Month) for December 2025; promoted to Service Technician in April 2026

Member Services Representative Oct 2024 – Aug 2025
Planet Fitness *Chantilly, VA*
- Handled front-desk operations including member check-in, enrollment, billing transactions, and cancellations

Skills

Languages: Python, C, C++, Rust, JavaScript, TypeScript
Technologies: Machine learning (neural networks, reinforcement learning, LLM inference), OpenCV, Flask, Tauri, Svelte, Docker, Google Cloud Run, Gemma/Gemini APIs, Linux (Ubuntu), PC hardware diagnostics
Tools: Git, GitHub Actions, Visual Studio, VS Code, PyCharm, Cargo
Certifications: Apple, HP, Dell, Lenovo (NGSP/DOA), and ASUS authorized repair; CPR